

The Silent Thief: Pancytopenia due to Pernicious Anemia

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Introduction

Pancytopenia can stem from a variety of causes, including neoplastic conditions, splenic sequestration, nutritional deficiencies, autoimmune processes, and increased cellular destruction seen in disorders such as disseminated intravascular coagulation (DIC) and thrombotic thrombocytopenic purpura (TTP). B12 deficiency is relatively common, affecting up to 12% of adults under 60 and nearly 20% of those older. Due to vitamin B12's role as a cofactor in DNA methylation and nucleotide synthesis, its deficiency can cause intramedullary hemolysis, megaloblastic anemia and shortened RBC survival. In severe cases, intramedullary hemolysis due to B12 deficiency can present with findings of hemolytic anemia (e.g. high LDH, low haptoglobin, schistocytes) due to a phenomenon known as pseudo-thrombotic microangiopathy (pseudo-TMA). The most feared complication is subacute combined degeneration (SCD) secondary to increased levels of methylmalonic acid (MMA) which impairs myelin sheath synthesis in the spinal cord. High-risk groups for B12 deficiency particularly include individuals on strict vegan or vegetarian diets, as well as those with absorption issues (e.g. pernicious anemia, atrophic gastritis). Pernicious anemia, an autoimmune disorder caused by antibodies targeting intrinsic factor or parietal cells, leads to B12 malabsorption. Diagnosis is confirmed by anti-intrinsic factor and anti-parietal cell antibodies, which are 100% specific, though the sensitivity is only 50%, indicating a substantial risk of false negatives.

Case Presentation

A 59-year-old African American woman with no known medical history presented with 6 weeks of worsening fatigue, weakness, poor oral intake, and unintentional weight loss. She denied GI or GU bleeding and was not on any medications. Exam was notable for bilateral distal hyperpigmentation and proximal muscle weakness. Labs revealed pancytopenia with macrocytic anemia (Hb 6.9 g/dL, MCV 114 fL), ANC 350, platelets 21k, and schistocytes, macrocytes, and hypersegmented neutrophils on smear. Further workup showed elevated LDH, low haptoglobin, normal folate, undetectable B12, and normal retic. B12 deficiency was suspected, but hemolysis and thrombocytopenia raised concern for TTP. ADAMTS13 was normal, and MRI showed dorsal column hyperintensity consistent with subacute combined degeneration. Bone marrow biopsy revealed normocellular marrow with dyspoiesis, excluding malignancy. Despite a negative intrinsic factor antibody, pernicious anemia was confirmed with positive antiparietal cell antibody.



Figure 1 Hyperpigmentation of the palms and soles, a classic sign of chronic B12 deficiency

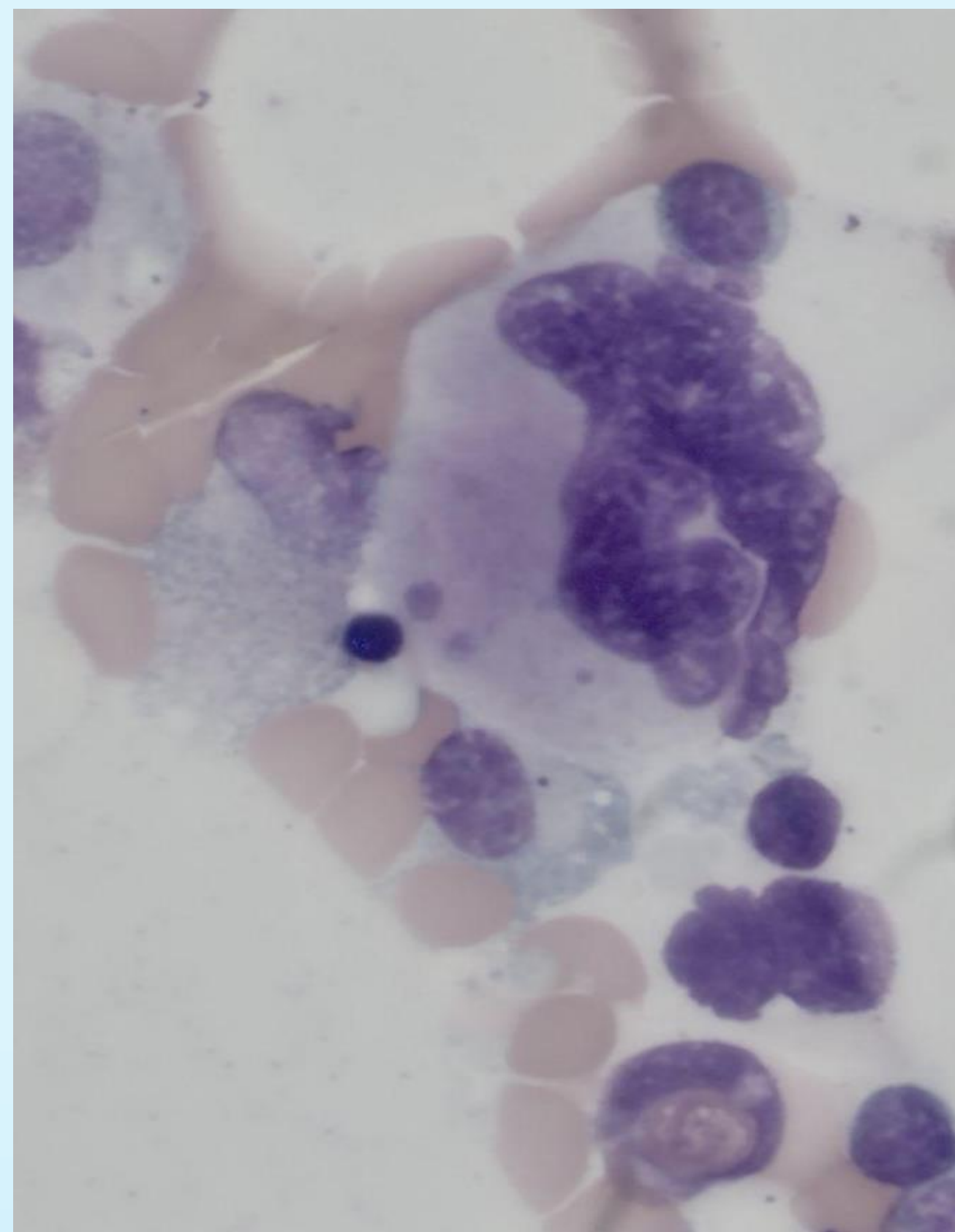


Figure 2 Bone marrow biopsy showing normocellular marrow with dysplasia and megaloblastic changes, supporting ineffective erythropoiesis

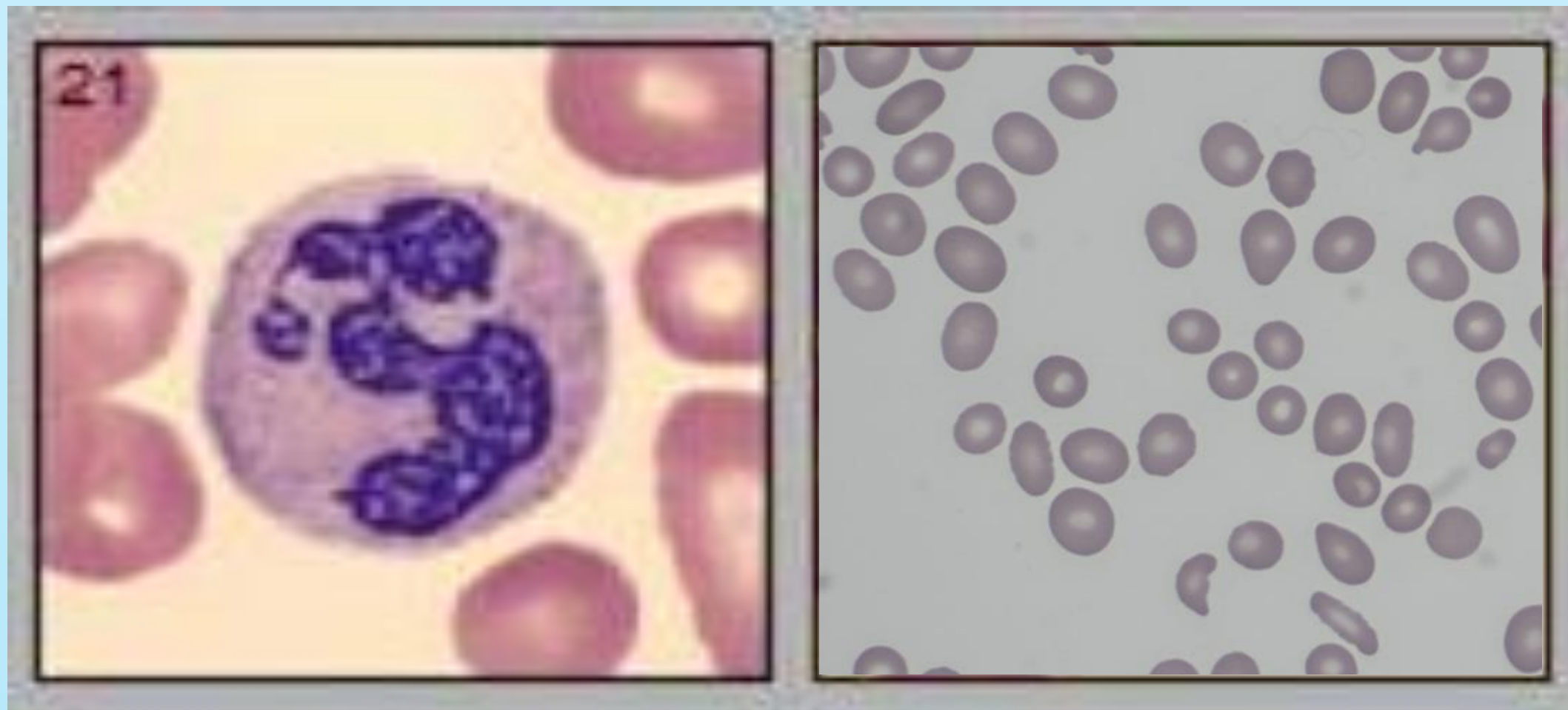


Figure 3 Peripheral smear demonstrating a hypersegmented neutrophil, anisopoikilocytosis, and macrocytosis

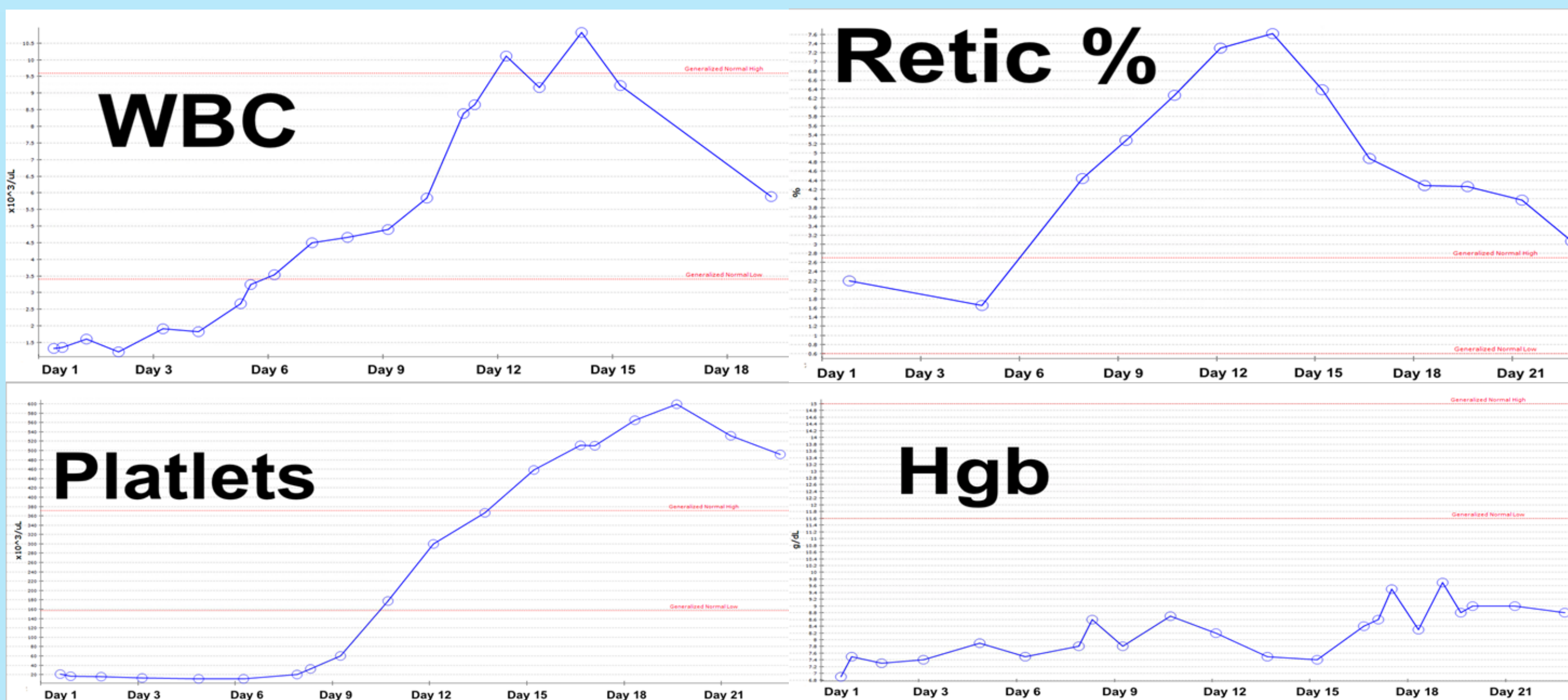


Figure 5 Trend of hematologic recovery following B12 supplementation

Conclusion

This case highlights several key points in the management of immunocompromised patients. First, low CMV viral loads do not exclude CMV chorioretinitis, and clinical judgment should guide empiric treatment in high-risk patients. Second, extending the induction phase of cryptococcal meningitis treatment may be necessary in severe cases, particularly in patients with persistently elevated ICP. Balancing antifungal efficacy with strategies to mitigate Amphotericin B toxicity is crucial, as nephrotoxicity and electrolyte imbalances can develop despite proactive measures. Finally, addressing social determinants of health is vital in patients with barriers to care, and collaboration with support services plays a key role in achieving long-term clinical stability.

References



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